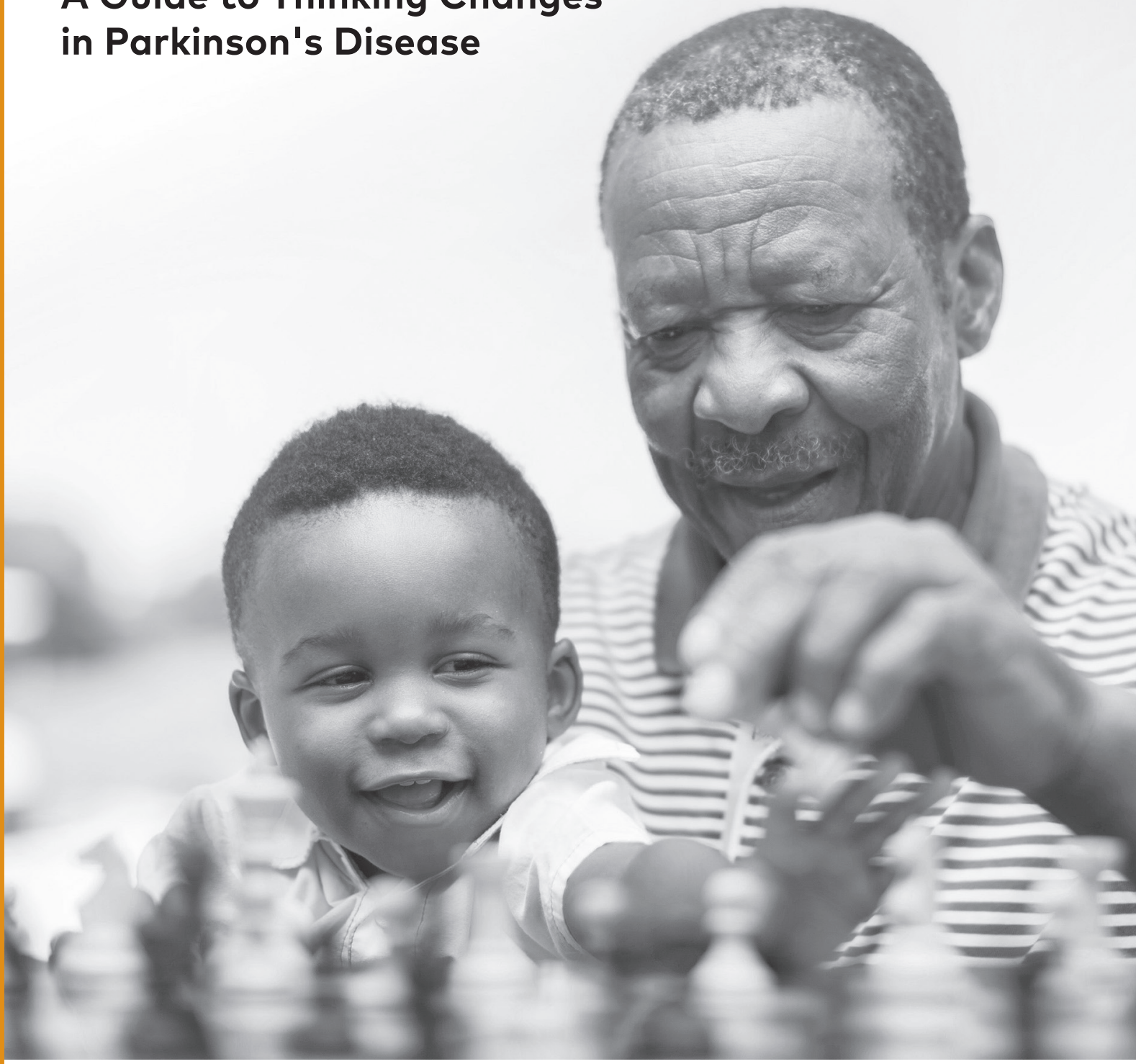
Cognition

A Guide to Thinking Changes in Parkinson's Disease



Parkinson's Foundation

About this book

Many people will experience some degree of thinking and memory changes (also called cognitive changes) while living with Parkinson's disease (PD), but not everyone will have significant symptoms.

This book is a practical guide to understanding how PD and other factors can affect cognition. It also outlines treatment options and provides tips and strategies for managing cognitive challenges.

The Parkinson's Foundation is here to help. As you review this book, direct any questions to our Helpline specialists. Call 1-800-4PD-INFO (1-800-473-4636) or email Helpline@Parkinson.org.

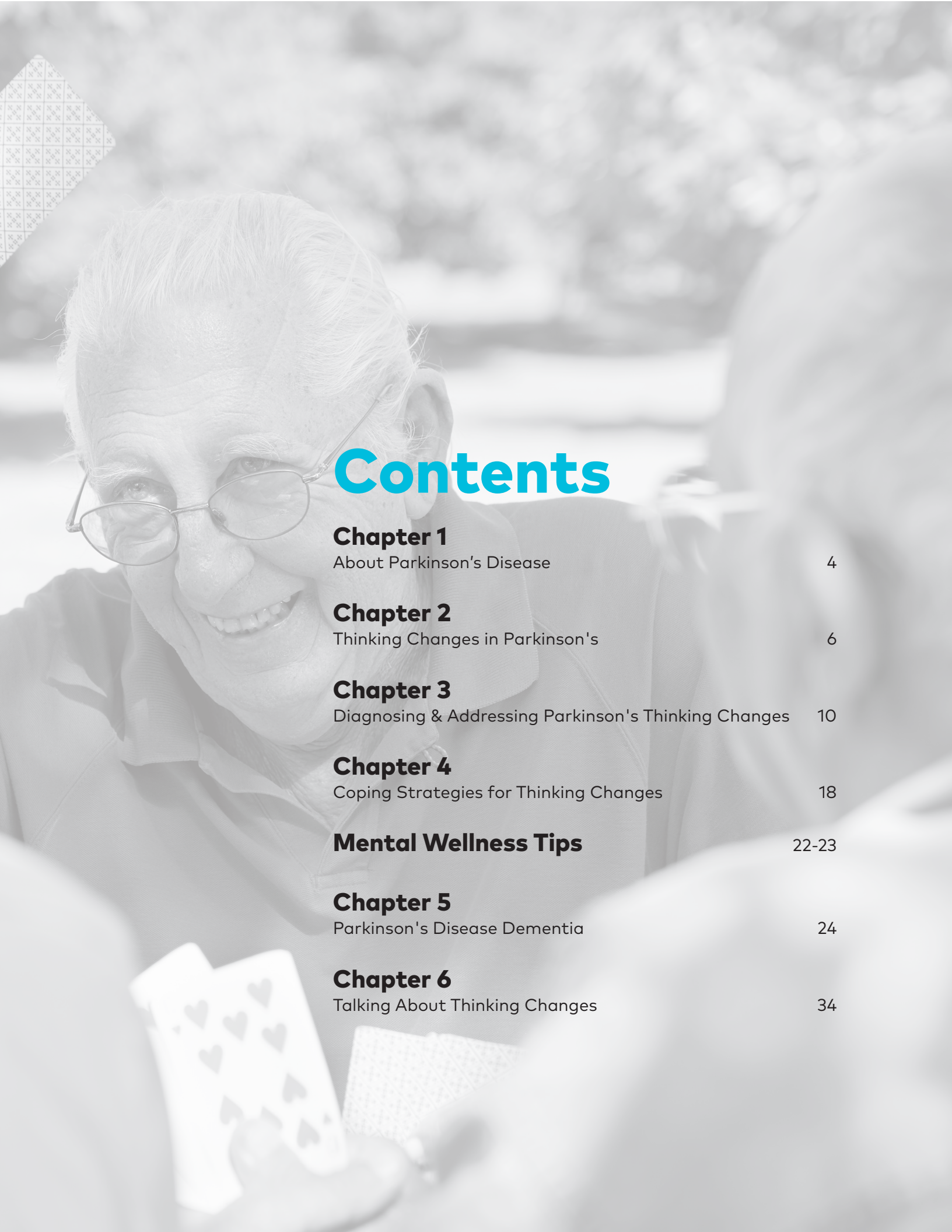
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1 About Parkinson's Disease

Let's review the basics:

- Parkinson's disease (PD) is a progressive neurodegenerative disorder that affects about 1 million people in the U.S. and 10 million people worldwide.
- Symptoms are varied and usually develop slowly over time.
- It is considered a movement disorder because it causes slow movements (bradykinesia) as well as other symptoms related to mobility, such as stiffness (rigidity), muscle cramping (dystonia) and problems with balance and walking.
- PD can also cause many non-movement symptoms, such as depression, sleep issues, constipation, fatigue and voice changes.
- Although there is no cure for Parkinson's at this time, many advances in treatment options continue to help people live better.



NOTE

While a PD diagnosis impacts each person differently, taking charge of the things you can and connecting with others in the Parkinson's community can empower you and improve your overall quality of life.

How is PD diagnosed?

Parkinson's is a "clinical" diagnosis. This means that a person's medical history, symptoms and a physical exam are used to make the diagnosis. There is not a single test that is commonly used to diagnose PD. However, brain imaging studies, such as an MRI or DaTscan, spinal fluid analysis or an alpha-synuclein skin biopsy test, may help make a Parkinson's diagnosis or rule out other conditions.

How does the brain change in PD?

Brain cells that produce a chemical called dopamine become damaged and gradually die — beginning years before a Parkinson's diagnosis. Dopamine is one of several chemicals the brain uses to communicate with the body. It is essential for movement but also plays a role in mood, behavior and thinking.

Researchers have discovered that Parkinson's is linked to Lewy bodies — abnormal clumps of the protein alpha-synuclein in parts of the midbrain and brain stem, upper part of the nose and the gut. The exact relationship between these proteins and the development and progression of PD is still unclear.

What causes PD?

Researchers believe that a combination of genetic and environmental factors contribute to the development of Parkinson's.

Genetic Factors

Dozens of gene mutations have been linked to Parkinson's, though researchers are still uncovering what role they play. Genetics cause about 10 to 15% of all cases of PD. However, even when someone has a PD-associated gene mutation, the likelihood of developing the disease can be low.

The Parkinson's Foundation offers genetic testing and counseling at no cost for those diagnosed with PD through its global genetics study. Understanding the connection between Parkinson's and genetics can help us understand how the disease develops and ultimately how it can be treated or cured.

Enroll now at Parkinson.org/PDGENEration.

Environmental Factors

A broad range of environmental exposures have been linked to PD including rural living, well water, manganese, pesticides and herbicides. Many people who are exposed to toxins never develop PD.

The extent to which environmental and genetic factors are involved in developing Parkinson's varies from person to person. The interactions can be quite complex.

2 Thinking Changes in Parkinson's

Parkinson's disease (PD) changes the brain, which can impact the whole body. While slowed movements, stiffness and tremor are among the more familiar symptoms, PD can also affect cognition — the way we think, remember, learn, use language and navigate space. Some people may not notice thinking and memory (cognitive) changes right away. Others can experience challenges even prior to a diagnosis. Typically, significant cognitive changes do not tend to develop until many years after diagnosis — if at all.

Quick Facts

- Thinking and memory changes are common in Parkinson's but can be mild, especially early in the course of the disease.
- In general, Parkinson's tends to affect memory less than other cognitive abilities, such as concentration, planning and multitasking.
- Coping with cognitive changes in PD can be challenging, but early recognition and treatment can make a significant difference in managing symptoms and improving quality of life.

Understanding Cognitive Changes

Cognitive changes in Parkinson's can be subtle, such as difficulty with multitasking or finding the right words. In some cases, people may question whether cognitive changes are part of the normal aging process or PD.

When thinking skills decline more than expected with age, but generally do not interfere with independence, this is called mild cognitive impairment (MCI).



NOTE

People with MCI can often compensate for cognitive changes by using strategies such as writing notes, using calendars or reminders and relying more on routine and habit to help manage daily tasks and activities. Small adjustments can make a big impact!

If cognitive changes start to limit the ability to manage everyday tasks and interfere with independence, they are generally described as dementia. People with Parkinson's disease dementia (PDD) experience some of the same challenges as those with MCI, but symptoms are more significant and may also include confusion, behavioral changes and some memory loss. Some people with PDD may face greater challenges, while others may be able to live independently with the right support.

Mild Cognitive Impairment Symptoms

Changes in one or two of the following areas is common in MCI:

Attention

Over time, people with PD may have problems with tasks that require extended focus or the ability to filter out distractions, such as balancing a checking account or having a conversation with a lot of background noise.

Slowed Thinking

Slowed thinking, known as bradyphrenia, can cause someone with PD to take longer to register and respond to information, participate in a conversation, remember details, follow directions and more.

Executive Function

These mental skills — including the ability to plan, organize, focus, manage time, multitask and solve problems — help us control our actions, follow through on tasks, achieve goals and make important decisions.

Language

Word-finding difficulties and trouble understanding complex or highly detailed language are among the challenges people with PD can have in conversations. Sometimes people will describe feeling like their brain and mouth are not coordinating.

Memory

A person may have problems with short-term memory and learning new information as PD progresses. For example, some people find it difficult to remember what they did earlier in the day or whether they took their pills. Using cues or hints can help jog memory. Often, with some time and patience, the word or memory will come back.

Although making new memories can be more difficult, remembering events from the past may be fairly easy.

Mood Changes

Depression, anxiety and a lack of motivation or interest, known as apathy, can all be common, underrecognized and undertreated symptoms in MCI. Apathy can impact energy, interfere with Parkinson's management and make it challenging for a person to stay engaged and inspired.

Address mood changes with your care team. Treatment can significantly improve well-being. Restful sleep, regular exercise, following a daily schedule and creating small weekly goals should also be part of an overall plan to boost emotional health.

Visuospatial Processing and Perception

Parkinson's can impact the ability to process visual information, such as judging distances, distinguishing contrast between similar colors, recognizing objects or figuring out spatial relationships. A person may have more trouble navigating their surroundings, including familiar places, and may get lost as PD advances. Visual problems associated with aging — such as macular degeneration or cataracts — can also contribute to visuospatial problems and may progress with time.



TIP

Thinking changes can make everyday tasks — such as paying bills, eating, dressing and participating in conversation — more difficult. If you or someone you care for are struggling with these kinds of activities, make an appointment to talk to your healthcare team. Speech-language pathologists (also called speech therapists) and occupational therapists can offer strategies to help.

"Once I knew more about how PD can impact thinking, I got better at noticing changes, talking to my doctors about them and finding ways to manage them."

-Julio, person with Parkinson's

Chapter Check In

What is one of your main takeaways from this chapter?

Are you experiencing any mood changes, such as being more irritable, feeling sad or having trouble concentrating?

Have you noticed any cognitive changes (whether you are a person with PD or a care partner)? Are these changes impacting your day-to-day activities?

3 Diagnosing & Addressing Parkinson's Thinking Changes

Researchers are making discoveries about how and why thinking and memory issues develop in Parkinson's disease (PD), but there is still much to uncover. Treating cognitive changes in PD can be complex and requires a thorough review of symptoms, medications and medical history. In some cases, cognitive symptoms may improve if there is a cause that can be identified and addressed effectively. While mild cognitive impairment and dementia cannot currently be cured, medications, adaptive strategies, a healthy diet, exercise and staying mentally and socially active can often help manage symptoms. Coping strategies, discussed in Chapter 4, play a key role in maintaining quality of life for people with cognitive changes.

Quick Facts

- Alongside PD, factors such as aging, medication side effects, infections, depression and sleep issues can impact thinking abilities.
- Recognizing cognitive changes early allows for better management and improved quality of life.
- Collaboration among primary care doctors, neurologists and other specialists like occupational therapists and speech-language pathologists can help support the management of cognitive changes in Parkinson's.

Building a Supportive Care Team

Your primary care provider and neurologist play important roles in diagnosing and managing cognitive changes. Your primary care provider typically handles general health issues that might impact cognition, while your neurologist focuses on PD-specific treatment and medication adjustments, including "cognitive enhancer" medications. Both can recommend resources and refer you to other cognitive specialists if needed. To help your care team work well together, remind them to review all appointment notes and recommendations.

Other healthcare professionals can play important roles in assessing and navigating thinking and memory issues with PD, including:

- **Neuropsychologists** evaluate memory and thinking skills to create personalized strategies that help manage cognitive challenges and improve everyday abilities.
- **Psychiatrists** can help diagnose cognitive and mood changes and recommend medications to manage symptoms.
- **Speech-language pathologists,** also called speech therapists, help evaluate and improve cognitive-communication skills, including speaking, organizing and understanding language, attention, memory and executive function.
- **Occupational therapists** specialize in improving the ability to do everyday tasks, like organizing medication schedules and modifying the home environment to make it safer and more manageable for people experiencing cognitive changes.
- **Social workers** can identify resources in the community for adapting to cognitive changes, including in-home care services, mental health counselors and support groups, as well as offer guidance around coping and planning ahead.

Our ability to think is a big part of who we are. Talking about cognitive changes can be difficult or even feel scary, but open dialogue can lead to timely diagnosis and personalized strategies. These can help maintain independence and mental well-being.

If you notice any thinking or memory changes, consider the following questions and let your care team know:

- What am I having trouble doing or remembering?
- How frequently do these symptoms occur?
- Are there specific triggers (time of day, noise, being alone)?
- Has my thinking or memory worsened or improved with any medication changes?
- Do I experience foggy thinking when my PD medications "wear off"?
- Is there something else, like an illness?
- Was the change sudden or gradual?



NOTE

A family member or close friend can likely add observations about cognitive symptoms that will be useful for the healthcare team.

Diagnosing Cognitive Changes

Figuring out whether cognitive changes are related to Parkinson's, aging or other factors can be challenging. Often, healthcare professionals will start by looking for causes of cognitive changes that could possibly reversed. A thorough evaluation of cognition can involve:

Medical History Review

Providing your medical history, family history, medications and existing health conditions can help identify factors that may be affecting your thinking.

Neurological Examination

A neurologist evaluates symptoms (including cognitive function) to determine if they are related to Parkinson's or something else.

Neuropsychological Testing

Tests like the Montreal Cognitive Assessment (MoCA) and Mini-Mental State Examination (MMSE) look at memory, attention and other thinking skills. These tests help find specific problem areas and guide treatment. Establishing a cognitive baseline early on can help track changes over time and adapt strategies.

"I wanted to rule out the possibility that I was somehow imagining everything. So, I directly sought out a neuropsychologist and went through the whole battery of cognitive testing. I just wanted to know."

-Lisa, person with Parkinson's

Some tests are given in the primary care, neurology or rehabilitation settings by doctors, physician assistants, nurses, speech-language pathologists or trained assistants. Others need a referral to a neuropsychologist or neuropsychiatrist.

Lab Tests and Imaging

Blood tests check for metabolic disorders, vitamin deficiencies and infections. Scans like an MRI or CT can show if there are problems in the brain affecting how you think.

Addressing Factors That Affect Cognitive Health

Care teams often start by reviewing medications and addressing other factors that might impact cognition.

1. Review Medication Side Effects

Certain types of medications can cause or worsen cognitive symptoms in Parkinson's. These include anticholinergic drugs often found in sleep aids, allergy medications and some bladder medications, benzodiazepines and opioids. Changes in medications can sometimes contribute to delirium — a sudden state of confusion and disorientation.

Medications prescribed to manage movement symptoms in PD can also affect thinking. When addressing concerns about thinking and memory, doctors typically stop Parkinson's medications in this order: first anticholinergics, then MAO-B inhibitors, amantadine, dopamine agonists and COMT inhibitors. Finally, they decrease the dose of levodopa.

Your doctor may recommend reducing the dosage of a specific medication or discontinuing it altogether to see if symptoms improve, before exploring other treatment options.

2. Check for Other Issues

Mood Disorders
Depression and anxiety affect many people with PD and often go undiagnosed or untreated. These conditions can have cognitive symptoms, such as slower thinking, difficulty concentrating and memory troubles. Treatment typically involves counseling, psychotherapy and medications.

Sleep Problems
Sleep disturbances are common in PD and can significantly worsen cognitive symptoms. Issues include:

- Trouble falling or staying asleep: Often worsened by movement symptoms, particularly when medications wear off.
- REM behavior disorder (RBD), where people physically act out dreams during the rapid eye movement phase of the sleep cycle.
- Frequent nighttime bathroom trips.
- Nighttime restlessness and vivid dreams.
- Sleep apnea, marked by snoring and pauses in breathing while sleeping.

Managing sleep problems is important for thinking, memory and overall well-being. Approaches include:

- Sleep hygiene: Establishing a consistent bedtime routine, limiting or shortening daytime naps, ensuring a comfortable sleep environment, getting exposure to daylight, staying physically active and avoiding alcohol, nicotine, caffeine and heavy meals before bed.
- Medication adjustments: Parkinson's medication adjustments or sleep aids may help improve sleep quality. Talk to your doctor before trying sleep aids. Some can worsen cognition. Melatonin does not worsen cognition, but it is sometimes combined with other agents, so check with your doctor first.

For more information about managing sleep issues, visit Parkinson.org or contact the Parkinson's Foundation Helpline at **1-800-4PD-INFO (1-800-473-4636)** or **Helpline@Parkinson.org**.

Fatigue
Fatigue is a common symptom in PD that can contribute to "brain fog" and affect concentration and memory. Treatment options include adjusting medications, addressing depression, improving sleep habits and regular exercise.

Low Blood Pressure
Orthostatic hypotension (OH) causes a sudden drop in blood pressure when standing up, which can cause symptoms like dizziness and fainting. Regular screenings and reporting symptoms to your doctor are important.

Dehydration
Not drinking enough liquids during the day, especially water, can lead to dehydration, which can cause slowed thinking, poor attention, increased confusion and lead to higher risk of falls, urinary tract Infections and constipation.

Diabetes
Untreated diabetes can worsen thinking and memory issues. Regular cognitive evaluations are recommended if you have diabetes or are at risk.

Stroke
Stroke can affect cognition by damaging parts of the brain. Managing stroke risk factors through lifestyle changes and medications is important for cognitive health.

Infections
Urinary tract infections, pneumonia and other infections can cause abrupt changes in thinking and memory. Seeking prompt medical attention is important if you notice sudden cognitive changes.

Hospital Stays
During hospitalization, medication side effects, infections and unfamiliar environments can lead to confusion or delirium.

Learn more about hospital safety and order your free Hospital Safety Guide at Parkinson.org/HospitalSafety.

Unfamiliar Environments
In addition to hospitals, any unfamiliar setting (staying with family, traveling in a new city, airports, skilled nursing facilities, etc.) can potentially increase anxiety, confusion and cause disorientation.



NOTE

Sudden cognitive changes or confusion over a short period (hours, days, weeks) are unusual in PD and should prompt evaluation for other medical issues, such as a urinary tract infection, stroke or medication side effects — which are often treatable.

Call our Helpline at **1-800-4PD-INFO (1-800-473-4636)** or visit Parkinson.org/Library to learn more about medication, sleep, fatigue, low blood pressure or hospital safety. Download *Medications: A Treatment Guide to Parkinson's Disease*.

Tips for Addressing Cognitive Changes

In addition to addressing factors like medication side effects and other health issues, the following tips can help support cognitive health.

1. Review Medication Options

Certain medications may help manage cognitive symptoms in PD, including:

- Selective serotonin reuptake inhibitors (SSRIs), typically used to treat depression, may also be helpful if thinking problems are linked to mood or anxiety issues.
- Adjusting medications that boost dopamine may help improve thinking by reducing symptoms, which can lessen physical discomfort and distractions.
- Some medications used for Alzheimer's disease may help improve some cognitive symptoms for people with Parkinson's disease dementia (PDD). Learn more in Chapter 5 (page X).

2. Exercise Regularly

Exercise is not just about moving your body; it is a powerful tool for maintaining and improving thinking and memory in Parkinson's. Studies show that regular exercise can make it easier to concentrate, process information and think clearly.

Parkinson's Foundation research shows that at least 2.5 hours of exercise a week can improve quality of life.

Cognitive Benefits of Exercise

We have known for years that exercise improves muscle strength, flexibility, bone density and heart health. New research shows that when you take up a new exercise, your brain is also activated — not just your muscles. This process of teaching your brain a new pattern or movement is called neuroplasticity. Animal studies suggest that this process may have many benefits in Parkinson's, including:

- Helping brain cells use dopamine more effectively.
- Promoting blood vessel growth, which helps brain cells get the oxygen and nutrients they need to stay healthy and work well.
- Releasing brain growth factor — a substance that helps brain cells make new connections.
- Growing new brain cells.
- Supporting the immune system

Aerobic vs. Skilled-based Exercises

Research shows that aerobic exercise, like treadmill training, dance and stationary bike, can improve our thinking abilities as we age and may reduce some symptoms of mild cognitive impairment. Skill-based exercises or "goal-oriented movements," including dancing or yoga, can help address movement symptoms, such as walking, posture and balance problems.

Activities that combine aerobic exercise and skill include:

- Playing sports such as boxing, Ping-Pong and pickleball.
- Cycling on a stationary bike, changing the intensity or navigating a virtual obstacle course.
- Walking quickly while chatting with a friend.
- Attending group fitness classes with memory games or coordination exercises.
- Doing outdoor activities like hiking that involve navigating trails

Physical and occupational therapists can help create a personalized exercise plan that's safe and effective for you.

"Finding the right exercise routine has been a game-changer for my overall well-being. It has not just helped with my movement symptoms, but also improved my sleep and mood. The biggest surprise has been how much clearer my mind feels now that I'm exercising regularly."

-Sanjay, person with Parkinson's

For more information on the benefits of exercise and guidance on exercises you can do, visit Parkinson.org/Exercise. Order your free copy of the publication *Fitness Counts* at Parkinson.org/Store or contact our Helpline at **1-800-4PD-INFO (1-800-473-4636)** or **Helpline@Parkinson.org**.

3. Strengthen Cognitive Abilities

Cognitive training uses repetitive, guided practice to possibly strengthen thinking skills. Research suggests that depending on the specific cognitive challenge, some people may experience improvements in their ability to remember, focus, plan and solve problems.

Professionals who oversee cognitive training include neuropsychologists, cognitive therapists, occupational therapists or speech-language pathologists.

"Working with a neuropsychologist was very helpful to me. She was able to point out mental and physical exercises I could do. I felt like that if I was consistent in doing those, they would become sort of implanted in my thinking process and I would become more capable of meeting the cognitive challenges when they came, and that's true."

-Johnny, person with Parkinson's

Chapter Check In

What is one of your main takeaways from this chapter?

Who can I add to my care team who can help to monitor and address my cognition?

What is my current biggest question about cognition that I can ask at my next neurologist appointment?

4 Coping Strategies for Thinking Changes

If you are experiencing thinking challenges that are interfering with day-to-day life and relationships, there may be ways to adapt your surroundings and strategies that can help you with these challenges.

Quick Facts

- Creating routines, using reminders, organizing tasks and reducing distractions can make daily life easier and more manageable.
- Enjoying hobbies, learning new skills and participating in social activities supports thinking, memory and overall well-being.
- Mindfulness practice, participating in support groups and seeking professional guidance can help manage stress and navigate cognitive changes.

Navigating Daily Tasks with Cognitive Changes

Use and adapt these tips to help manage your individual needs and challenges:

- Establish routines. It can make building habits easier. Adding a new task to an already existing habit is especially successful. People experiencing apathy or low motivation — a common PD cognitive symptom — may need family and friends to help schedule and follow through with activities.
- Use reminders. Calendars, smartphone reminders, alarms, Post-its, checklists and other visual and audio cues to remember important details during the day — such as meeting up with a friend for dinner, taking medications on time and bringing your wallet when leaving home.

- Develop systems. For example, organize pills into a pill box once a week or arrange your kitchen so all the items you need to complete a task, like making coffee, are in one area.
- Cue yourself to use your assistive devices. For example, if you use a walker for bathroom visits during the night, leave it close to your bed.
- Focus on one thing at a time, especially if it is challenging to multitask or stay focused.
- Divide your tasks into individual steps that are easier to complete.
- Avoid loud rooms and distractions when possible. Turn off the television or radio while you are having a conversation. Keep rooms well-lit if you experience problems with visual perception. Learn more about assistive devices and review our home safety checklist at Parkinson.org/HomeSafety.



NOTE

Speech-language pathologists, occupational therapists and neuropsychologists can help you learn and use these strategies effectively to fit your unique goals and cognitive abilities.

Staying Socially and Mentally Active

Studies suggest that staying mentally and socially active not only supports cognitive skills but can also enhance overall quality of life for people with Parkinson's, and care partners.

Engaging with others helps build community, create purpose and reduce feelings of isolation and loneliness. Consider joining support groups, discussion groups, creative classes or volunteering. Participating in clubs, hobbies and spending time with friends and family can also support mental and emotional well-being.

Mental engagement stimulates cognitive function and promotes mental sharpness, potentially slowing cognitive decline and boosting brain health. Do puzzles, brain games and learn new skills. Read, attend lectures and pursue interests to keep the mind active and engaged.

Managing Stress

Experiencing cognitive changes or anticipating potential changes can be overwhelming, stressful and sometimes lead to anxiety. Mindfulness techniques like deep breathing exercises, meditation or yoga can help calm your mind and boost overall well-being. "Mindfulness" also includes other activities that bring joy and relaxation, such as listening to music, reading, taking a walk, spending time outdoors or simply being with friends and family.

Another effective way to reduce stress is by attending support groups, where you can find a safe space to connect with others who understand your challenges and share practical tips.

For more structured support, psychologists, counselors, clinical social workers, marriage and family therapists and other mental health professionals offer tailored strategies to lessen stress and build coping skills for managing cognitive changes. For example, a therapist trained in cognitive behavioral therapy can help you develop tools to change behaviors and thoughts that impact your emotional health.

Therapists listen to your experiences, concerns and feelings and provide nonjudgmental feedback that can benefit both people living with PD and care partners. There are many places you can look for a counselor:

- Ask members of your support group or health care team for a referral
- Call your local Area Agency on Aging
- Search therapist directories (such as Psychology Today or Inclusive Therapists)
- Contact our Helpline at **1-800-4PD-INFO (1-800-473-4636)** or **Helpline@Parkinson.org**



TIP

Living alone? It is especially important to build a strong support network. Family, friends, neighbors, support groups and healthcare professionals can help you stay active, assist with tasks and help manage emergencies.

Chapter Check In

How have you adjusted your daily routines to manage cognitive challenges, and what impact has this had on your daily life?

How do you stay active socially and mentally? List some new activities or hobbies you want to explore.

Have you found a support group where you feel comfortable and encouraged? What kinds of practical strategies have you learned for navigating day-to-day cognitive changes?

How do you manage stress? Are there new techniques you would like to explore?

Mental Wellness Tips

Self-care, creative strategies and staying social can help you maintain your mental well-being while coping with thinking changes. Consider these tips:

- Our memories and the way we think form important parts of our identities. Give yourself permission to feel difficult emotions. Experiencing cognitive change can bring up grief or fear. Recognize and honor your feelings around these changes.

- Share any cognitive challenges you are having with your healthcare team so they can support you and offer resources.

- Lighten your load. Ask for and accept help — whether with transportation, grocery shopping, getting up from a chair, staying on top of medical appointments or making your home safer. Even though it can be difficult, being open to assistance allows you to focus your physical and cognitive energy on other important tasks and activities.

"Last year it became too difficult for me to put my medications into the medication organizer. It wasn't like I'd make terrible mistakes, but it would take me a long time. I would put them all in and wonder, 'Did I just put all of those in nighttime boxes when they should have been daytime?' It's just easier to have my husband do it and one of the benefits of that is that it has actually brought him into my care in a way that he wasn't before."

-Lisa, person living with Parkinson's

- Lessen your stress. Studies show that stress can worsen PD movement and non-movement symptoms, including executive function. Physical exercise, guided relaxation techniques and mindfulness boost mood and brain power and are linked to symptom improvement. Get involved at Parkinson.org/MindfulnessMondays and Parkinson.org/FitnessFridays.

- Figure out what works for you. For example, if you love to cook but have trouble remembering which ingredient is next, consider lining up all the ingredients you need in order and putting them away as you go. Physical, occupational and speech therapists can help problem-solve safe and effective ways to continue participating in the activities you love.

- Stay connected. Strong social connections can support cognitive health.

  - Cultivate relationships with friends, family and community members
  - Consider joining a support group to connect with others who understand your experience.
  - Contact our Helpline at **1-800-4PD-INFO (1-800-473-4636)** or **Helpline@Parkinson.org** to find a local group or visit PD Conversations, our online community at PDConversations.org.

Mental Wellness Tips for Care Partners

Supporting someone with cognitive changes can make self-care seem overwhelming. These tips are designed to help care partners stay healthy and resilient:

- Remember to take care of yourself, too. Your physical and mental well-being is just as important as the health of the person you are caring for.
- Be kind to yourself and acknowledge your right to feel emotionally off-balance. Remember you are experiencing normal reactions to challenging circumstances.
- Build in regular breaks to relax, socialize and exercise. It is important to maintain your interests and passions throughout your loved one's journey with PD.
- Consider seeking outside help, whether from a family member or a paid caregiver. Starting early can ease the transition, especially if cognitive changes become more significant.
- Use a journal or other creative outlet to express your feelings.
- Determine your physical and emotional limits and get help as needed. You cannot meet every need.
- Talk to someone. This can be a friend or family member, but a professional counselor offers objective insights and advice.

"I know it's difficult for my mom when I bring up her schedule mix-ups, and I can see she's scared and frustrated. But I needed support too, so I started going to a care partner support group. Talking with others who understand what I'm going through has been a huge help."

-Grant, care partner

For more information and additional tips, visit Parkinson.org/Caregiving.



Reach out to the Parkinson's Foundation Helpline for answers to your PD questions.

Call **1.800.4PD.INFO (1.800.473.4636)** or email **Helpline@Parkinson.org.**

5 Parkinson's Disease Dementia

As Parkinson's disease (PD) progresses, thinking and memory changes may become more noticeable. If these changes make managing daily activities alone difficult, it could be a sign of dementia — a condition marked by a decline in cognitive abilities that impacts daily life and independence. Dementia is not a specific disease, but instead a group of cognitive symptoms caused by different conditions. It evolves over time and impacts independence to varying degrees.

Quick Facts

- Parkinson's disease dementia mostly involves a decline in executive functions, such as planning and decision-making, unlike Alzheimer's disease, which is known for memory loss and difficulty with language.
- Providing care for someone with Parkinson's disease dementia can bring additional challenges, increasing the risk of burnout for family and care partners.
- Dementia can bring significant changes to your life and a variety of questions and emotions. With support, understanding and effective management strategies, people can cope better and lead fulfilling lives.

Parkinson's Disease Dementia

As Parkinson's movement symptoms progress, cognitive symptoms may worsen, possibly developing into dementia. When this happens, it is called Parkinson's disease dementia (PDD). Not everyone with PD will develop mild cognitive impairment or dementia.

Dementia Risk Factors

Certain risk factors are associated with dementia for people with PD. These include:

- Older age

- Longer duration of PD or developing it at an older age
- Family history of dementia
- Significant movement symptoms

Parkinson's Disease Dementia Symptoms

- Confusion
- Difficulty with reasoning and judgment
- Trouble with planning, problem-solving and other executive functions
- Hallucinations (seeing things that are not there)
- Delusions (holding strong, irrational beliefs despite evidence against them)
- Mood changes, such as apathy (reduced motivation), depression or anxiety
- Memory loss (although information can often be recalled with prompting or cueing)



NOTE

Conditions like urinary tract infections or medication side effects can cause sudden confusion, known as delirium. Delirium is reversible. If you experience a sudden change in thinking, let your doctor know right away.

Dementia with Lewy Bodies

If someone is experiencing dementia before or within the first year of movement symptoms, it is important to consider a diagnosis of dementia with Lewy bodies (DLB) — especially if they experience hallucinations. Another thing to note is that people with DLB might not benefit as much from levodopa therapy.

DLB is part of a group of conditions called atypical parkinsonism. Visit Parkinson.org/Parkinsonism to learn more about parkinsonism disorders such as multiple system atrophy, progressive supranuclear palsy and corticobasal degeneration. These conditions cause symptoms that resemble Parkinson's disease, but often progress quicker than PD and may have additional symptoms.

If your diagnosis is not clear, make an appointment with a movement disorders specialist (a neurologist with training in PD, DLB and more) to determine the cause of your symptoms. Some movement disorders specialists offer virtual appointments. Contact the Parkinson's Foundation to find a specialist near you. Call **1-800-4PD-INFO (1-800-473-4636)** or email **Helpline@Parkinson.org**.

Alzheimer's Disease vs. Parkinson's Disease Dementia

Because Alzheimer's disease (AD) is the most common type of dementia, many people assume that dementia is always caused by Alzheimer's. This is not the case. Alzheimer's and Parkinson's are both progressive neurodegenerative disorders that usually occur later in life, but they are different:

- **Alzheimer's** is a memory disease that does not usually start with movement symptoms. People with Alzheimer's have language difficulties earlier than people with PD and do not form new memories.

- **Parkinson's disease dementia** occurs when movement symptoms have been present for at least a year (more often more) before the dementia symptoms begin. People with PDD have more ability than those with Alzheimer's to remember information and events with cues. For example, the hint "flower" could help a person with PDD remember the word "rose."

Alzheimer's and Parkinson's can sometimes occur together.

Overview of Types of Dementia

Alzheimer's disease (AD)	Parkinson's Disease Dementia (PDD)	Dementia with Lewy Bodies (DLB)
What is it?		
The most common neurodegenerative disorder and leading cause of dementia	Can develop in people with Parkinson's disease years after a diagnosis and initial movement symptoms	Significant cognitive changes appear early on, often alongside movement symptoms
How common is it?		
60-80% of dementia cases	Together with DLB, PDD is the second most common form of dementia caused by neurodegeneration	Together with PDD, DLB is the second most common form of dementia caused by neurodegeneration
Risk factors		
Age, family history, genetics	Age, PD beginning at older age, longer PD duration, family history of dementia, severe movement symptoms	Age, male sex, family history of PD genetics

Alzheimer's disease (AD)	Parkinson's Disease Dementia (PDD)	Dementia with Lewy Bodies (DLB)
Most common symptoms		
• Memory loss gets in the way of everyday life • Trouble making new memories • Language issues – difficulty finding the right words when talking or writing	• Movement symptoms such as slowness of movement, difficulty walking and rigidity Problems with: • Short-term memory • Executive function: Maintaining attention, staying focused, processing information, problem-solving, planning and multitasking • Understanding and moving around in space (visuospatial perception) • Seeing or believing things that are not real (hallucinations and delusions)	• Movement symptoms such as slowness of movement, difficulty walking and rigidity • Dementia before or around the same time as movement issues start • Thinking and memory problems increase over time • Changes in alertness and attention (cognitive fluctuations) • Seeing or believing things that are not real (hallucinations and delusions)
Memory differences		
Cues do not usually help with memory	Cueing can help with remembering people, objects and events	Memory changes fall between Alzheimer's and PD



NOTE

Many conditions can lead to dementia, including other neurodegenerative diseases.

Managing & Treating Parkinson's Disease Dementia

Many of the treatment and management strategies outlined in Chapter 3 can also help people with PDD navigate thinking and memory changes. Below are additional strategies and considerations:

Routine and environment strategies
Establishing regular schedules for medication, sleep, meals and exercise can improve daily life. Setting up your home for comfort and reducing stress and confusion might involve decluttering, improving lighting and balancing quiet times with activities. Occupational therapists specialize in these strategies.

Meaningful activities
Focusing on activities that highlight strengths and interests may be more beneficial than traditional memory games for individuals with dementia.

Social engagement
Connecting with community resources such as dementia support groups, memory cafes and creative arts programs tailored for people with dementia is essential for maintaining well-being.

Medications for advanced cognitive symptoms
For Parkinson's disease dementia, a medication called rivastigmine, as well as others commonly used for Alzheimer's, such as donepezil and galantamine, may offer some benefits. These medications, known as acetylcholinesterase inhibitors, boost acetylcholine by blocking an enzyme that breaks it down, which may improve brain function.

Managing Hallucinations and Delusions
If hallucinations in PDD are not distressing, they may not require treatment, although it is still important to discuss them with your neurology team. Validating the experience and changing the topic can sometimes manage frustrations around hallucinations or delusions. For those needing medical treatment, atypical antipsychotics like clozapine, quetiapine and pimavanserin are considered safe for Parkinson's.

Regular Exercise
Regular exercise not only enhances mobility and mood but also offers potential benefits for managing symptoms of dementia.

Caring for a Person with Dementia

Caring for someone with dementia is a filled with rewarding moments and challenges. There are ups and downs, good days and bad. The experience of giving support is different for everyone, depending on the person's dementia symptoms, their relationship and the support and resources they have. These tips can help:

Communication

Practice patience, slow down and allow enough time for the person to process what you are saying and respond. Dementia can cause difficulty processing information and expressing thoughts and emotions. These tips can improve communication and minimize misunderstanding:

- Speak slowly and use simple sentences when communicating — this reduces pressure and gives a person time to process the information.
- Ask one question at a time and wait for an answer. Ask "either/or" questions. Instead of "What would you like for lunch?" try asking, "Would you like soup or a sandwich for lunch?"
- Limit distractions as you try to accomplish the daily routine. Make sure to turn the TV or radio off before asking the person to do something.
- Make eye contact and ensure you have the person's attention before speaking.
- Try not to argue. Instead, try asking questions to better understand their perspective. If the person seems stuck on a certain idea, validate their feelings and gently guide them to another topic or activity.
- Stay calm. If you are getting frustrated, take deep breaths or briefly step away. Calming your body and mind can allow you to think and respond more clearly.
- Consider the "why" behind behavioral changes like agitation. Is the person hungry, thirsty, tired, in pain, frustrated, lonely or bored?

Listen, validate and redirect rather than trying to correct the person. If someone is experiencing a hallucination, delusion or appears confused, denying their reality may cause fear, frustration or confusion.

Safety

Medication

For those with memory issues, it is not safe to handle medications alone. Family members can help by using a daily or weekly medication box along with setting alarms or a more high-tech option like an electronic pill dispenser that gives medications at the right time. As memory problems worsen, the care partner may need to take over medication management.

Eating

Dementia can affect eating habits. While using utensils might become harder, eating finger foods, like soft sandwiches, can be easier. Try different textures — smooth or crunchy — to help you find what foods your loved one is most comfortable eating. A speech-language pathologist can help with this.

Finances

As dementia progresses, family members or another trusted individual will need to ensure bills are paid accurately and on time and protect against financial abuse.

Driving
When thinking problems worsen, driving becomes unsafe. Some people with dementia lack the insight to stop driving, so care partners or doctors sometimes need to step in and say it is time to retire the keys. Learn more in Chapter 6.

Home environment
Removing tripping hazards, installing handrails and grab bars, improving lighting, getting an evaluation for mobility aids if needed and organizing and labeling items are just some of the modifications that can improve safety in the home.

Independence
Ultimately the decision to leave someone with dementia alone should be based on a thorough evaluation of their needs and risks. When dementia is moderate or more advanced, lack of judgment and impulsive behaviors can create a dangerous situation.

For more safety tips, visit Parkinson.org/HomeSafety.

Care Partner Burnout

Caring for someone with dementia can be demanding on care partners. Juggling medications, daily tasks and emotional ups and downs can be overwhelming. Care partners often put their own needs on the back burner, which can lead to burnout.

Signs of burnout include feeling exhausted all the time, losing interest in things you used to enjoy, not being able to sleep because of stress and even feeling angry or resentful towards the person you are caring for. Watch for these signs and seek support from others to take care of yourself while still providing the best care possible.

"I didn't even realize I had burnout until my daughter pointed it out to me ... you get to a point where you can't even take care of your needs because you are so overwhelmed. You've tried everything and nothing seems to be working."

-Marlene, care partner

You cannot pour from an empty cup. You must give yourself time and rest to fill up. Here are tips to help you find the energy to care for yourself:

- Each day, set aside time for relaxation, exercise or enjoyable activities that uplift you and build your resilience. Consistency is key, regardless of how much time you can dedicate.
- Ask family and friends for support. Having someone sit with your loved one while you take a break — whether for coffee, a walk or a self-care appointment — can help ensure safety while you care for yourself.
- Attend a PD or dementia care partner support group.
- Go to therapy to process your emotions and experiences and learn new coping skills.

Care Planning

- Make a list of important phone numbers and keep a copy with you in case of emergencies.
- Keep written track of appointments, tasks and medication schedules. Keep everyone on the same page — designate a corner in your kitchen to display the daily schedule and appointments.
- Create and follow a routine. If traveling or moving, try to maintain as close to a typical routine as possible to minimize stress.
- Give medications on time, as prescribed by the doctor.
- Use exercise, diet and rest to keep your loved one as healthy as possible.
- Work with a social worker who can recommend resources.
- Evaluate the need for outside help. Increased care needs, safety concerns, care partner burnout, a desire for more social interaction or declining care partner health could signal a need for outside help — either from family or friends, paid caregivers, an adult day care center or a care facility. To learn more about respite care, visit Parkinson.org/OutsideHelp.

Finding Support and Staying Informed

Support and education are essential for all care partners but especially when caring for someone experiencing dementia. In addition to your healthcare team, there are many resources for care partners.

Contact the Parkinson's Foundation Helpline at **1-800-4PD-INFO (1-800-473-4636)** for information about local and online resources, including dementia care specialists, support groups, care partner classes and professional caregivers.

Resources & Support

Dementia Specific

- Alzheimer's Association: alz.org; 1-800-272-3900
- Lewy Body Dementia Association: lbda.org; 1-800-539-9767
- Dementia Action Alliance: daanow.org

Local

- Area Agencies on Aging: Use the Eldercare Locator at eldercare.acl.gov or 1-800-677-1116 to find the closest one to you.

Private Care Management

- Aging Life Care Association: aginglifecare.org; 1-520-881-8008

Legal & Financial Advice

- National Academy of Elder Law Attorneys: naela.org

Chapter Check In

What parts of caregiving do you find most challenging or emotionally draining? What strategies do you use, or what new ones could you try to manage these challenges?

Check in with yourself. Rate your current level of burnout from 1-10.

What self-care practices do you use to promote your emotional and mental well-being?

Are there areas where you need help? Who can you contact to start exploring extra support?

6 Talking About Thinking Changes

Cognitive changes can be a sensitive topic. If you are experiencing these changes, you might be worried about judgment, burdening people in your support system or losing independence. Loved ones and care partners also find these conversations challenging. They may hesitate to discuss concerns, fearing it could cause emotional distress. However, these conversations can lead to better treatment, practical coping strategies and important preparation for the future.

Quick Facts

- Talking openly about thinking changes can lead to better support, treatment options and stronger relationships with loved ones.
- Parkinson's thinking changes can affect driving and work. Careful conversations can help address safety concerns and adapt to new challenges.
- Planning for legal, financial and medical matters helps make sure your wishes are respected and can make it easier to handle changes down the road.

Navigating Relationship Dynamics

Here are some helpful strategies to support healthy communication when someone is experiencing cognitive changes:

- Keep your conversations one-on-one or in small groups to lower stress and maintain better focus.
- Concentrate on one conversation and minimize distractions such as background noise.
- Plan discussions carefully — consider timing and who should be involved to allow sufficient time for meaningful dialogue.

- Talk face-to-face whenever possible rather than shouting from another room.
- Listen actively by giving each other ample time to respond and fully participate. Clarify understanding rather than assuming.
- Work on being open about your needs and challenges. Sharing openly can strengthen relationships, minimize misunderstandings, reduce stress and help you receive the support you need.
- Seek professional support from speech and occupational therapists for tailored communication strategies.
- Consult counselors or therapists for help navigating challenging conversations. Recognizing when assistance is needed can help maintain respectful and productive communication dynamics.
- Search therapist directories (such as Psychology Today or Inclusive Therapists) or contact our Helpline at **1-800-4PD-INFO (1-800-473-4636)** or **Helpline@Parkinson.org**.

Addressing Driving Concerns

Approaching the topic of driving can be particularly challenging. For many people, driving represents independence, freedom and a sense of control over their lives. However, waiting too long to broach the topic can compromise safety and make it harder to adjust to other ways of getting around.

Parkinson's can affect cognitive skills essential for safe driving, like decision-making, attention, multitasking, reaction time and spatial awareness. Movement symptoms such as slowed movements and muscle stiffness can also impact driving. Be aware that you may not realize you have a driving issue, so it may be important for you to ask for feedback from trusted friends, family members or your healthcare team.

1. Think about how you feel before, during and after driving (such as feeling anxious). Have you had any recent driving incidents, near misses or accidents?
2. Discuss your driving observations and concerns openly with family members, care partners and your healthcare team to work together to find practical solutions.
3. Consider a driving evaluation to receive a professional assessment of driving skills and tailored recommendations. Your doctor can also refer an occupational therapist with expertise in driving evaluation and rehabilitation or you can contact the Association for Driver Rehabilitation Specialists at 1-866-672-9466.



TIPS FOR FAMILY MEMBERS

- Try to keep your concerns specific. Instead of making general statements like "Your driving is terrible," give concrete examples such as, "Yesterday you went through a stop sign."
- Bring in help. Sometimes another family member, friend, healthcare professional or trusted acquaintance can offer input that is less emotionally charged and help facilitate a constructive conversation about driving concerns.

Managing Workplace Conversations

Deciding when to tell your employer about your condition is a personal choice. If you are experiencing thinking changes that affect your job performance, it may be beneficial to speak up sooner rather than later. However, disclosing could have risks, so it is often wise to seek guidance from someone outside of your workplace.

- Consider the impact on your job: Mild cognitive impairment (MCI) symptoms can make it difficult to perform your job effectively and may even jeopardize your employment if not disclosed and managed appropriately.

- These expert resources can help:

  - Job Accommodation Network (JAN) — a free consulting service provided by the U.S. Department of Labor offering answers about workplace accommodations and the Americans with Disabilities Act: askjan.org; 1-800-526-7234
  - Rehabilitation Services Administration (RSA) — operated by the U.S. Department of Education, RSA aids vocational rehabilitation agencies across the U.S. in helping people with disabilities maximize employment opportunities: rsa.ed.gov/about/states
  - Ticket to Work — Social Security Administration employment support services for Social Security disability beneficiaries who want to work: choosework.ssa.gov; 1-866-968-7842

Learn more. Read our Fact Sheet, Employment with Parkinson's Disease: Working It Out, at Parkinson.org/Employment.

Discussing Future Planning

Starting the planning process early will ensure that you have better control over important life decisions later on, including those involving finances, living arrangements and care.

Legal and Financial Planning:

- Complete legal documents such as a will, power of attorney for healthcare and finances and advance directives. Visit Parkinson.org/AdvancedCarePlan for more information.

- Consider designating a trusted friend, family member or professional as power of attorney to make decisions on your behalf if you become unable to do so yourself.
- Simplify finances by setting up automatic bill payments and organizing documents so others can understand and access information if necessary.
- Talk to an elder law attorney for specific legal and financial guidance. Visit the National Academy of Elder Law Attorneys at naela.org to find an elder law attorney near you. Local Area Agencies on Aging often have referral lists and offer additional resources and support. Visit eldercar.acl.gov to find your local agency.

"We were really afraid and so we put our heads in the sand for a while. But when we finally started looking at our [financial] options, it was less scary, and we were able to make some decisions."

-Marlene, care partner

Future Care Needs

- Discuss the role of family members and friends in providing ongoing care and assistance.
- Explore resources and support services available for care partners, including respite care options and support groups.
- Complete healthcare advance care directives as a tool to share wishes around medical care with families and healthcare professionals.
- Review healthcare coverage options and understand benefits and out-of-pocket costs related to in-home care, long-term care, mental health therapy and rehabilitation services and medical equipment and supplies.
- Research local assisted living facilities, skilled nursing facilities and other care options to be prepared if your needs change in the future.

Resources & Support

Legal & Financial Resources

- Medicare and Medicaid: medicare.gov and medicaid.gov
- Social Security programs: ssa.gov
- Family and Medical Leave Act: dol.gov/agencies/whd/fmla

Home Safety Resources

- Certified Aging-in-Place Specialist; call the National Association of Home Builders at 1-800-368-5242

Emergency Preparation Resources

- Parkinson's Foundation Blog Extreme Weather Emergency Preparedness; Parkinson.org/ExtremeWeather
- Local emergency management agency; fema.gov/locations

Communicating with Your Care Team

Communicating openly with your healthcare team can help you actively participate in managing your PD-related cognitive changes and other symptoms. Consider these tips to improve communication:

- Keep track of symptom changes, including any communication, thinking or memory issues, using a symptoms diary for 2-3 days before your appointment. Visit Parkinson.org/SymptomsDiary to download one.
- Write down 5-10 questions you have for your medical team and rank them in order of importance. Make sure you understand their answers. Most doctors are given a short amount of time to meet with each patient (15-60 minutes, depending on the appointment) so address the most important issues first.
- Have a family member or care partner accompany you to appointments to help recall details and communicate concerns. If you live alone, consider asking a trusted friend who is familiar with your symptoms. Some healthcare facilities offer patient advocates or social workers to provide extra support.
- Speak up about what matters most to you in your treatment plan.
- Ask about cognitive testing. Your doctor may refer you to a neuropsychologist.
- Ask about how to communicate in between appointments (phone calls, emails, patient portals) so you know how to easily relay updates or concerns to your healthcare team.

Chapter Check In

How can you start a conversation with your family or friends about any changes you're noticing in your thinking or memory?

What small steps can you take now to plan for your future needs, like legal documents or talking to a financial advisor?

What resources or organizations can you turn to for guidance about navigating difficult conversations?

What is the top Parkinson's concern or symptom you would bring up to your doctor or care team right now?

About the Parkinson's Foundation

The Parkinson's Foundation makes life better for people with Parkinson's disease by improving care and advancing research toward a cure. In everything we do, we build on the energy, experience and passion of our global Parkinson's community. A wealth of information about Parkinson's, events and resources are available at Parkinson.org.

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